NOAH FOUDIL

Junior Industrial Engineer
Electronics and Embedded Systems

VIEW PORTFOLIO → **ifloud2.github.io/Foudil_Noah_Portfolio**

Industrial electronics engineer, graduated from HELHA in 2026. I work at the hardware/software interface: board design (KiCad), firmware and embedded Linux, sensor integration and bench validation. I have taken projects from specification to test, in R&D, in industry and in teams. Used to taking a system end to end, from schematic to test, I am looking for my first engineering experience to bring this rigour to a team.

EXPERIENCE

R&D Intern, Embedded Electronics and UAV
DEC 2025 – MAR 2026

Yunlin University of Science and Technology (bioVLSI laboratory), Taiwan

- Integrated the perception chain of a GPS-denied indoor quadcopter: LD19 LiDAR and VL53L1X ToF combined with the EKF3 attitude of the ArduPilot controller over MAVLink
- Developed the on-board Python asyncio application (Raspberry Pi Zero 2 W): UART at 230,400 baud, MAVLink, ToF over I^2C, WebSocket, in 4 concurrent tasks
- Made the LiDAR link reliable (frame resynchronisation, CRC-8: 0.11 % rejected) and analysed a throughput loss (2,245 vs 4,500 pts/s) caused by scheduling
- Built the pseudo-3D reconstruction and its Three.js client; designed a connected-glove board in KiCad, validated on a breadboard
- Outcome: 4 of 5 targets met (≈ €269, 910 g, ≈ 7 min flight in a single trial)

R&D and Test-Bench Intern
SEP – OCT 2023

Hitachi Energy, Technology Center, Jumet

- Automated the processing of Fluke data-logger measurements (Excel/VBA tool taken over by engineers)
- Sized an RC filter in LTspice against supply ripple (HVDC tests, ≈ 5 kV)
- Designed the safety circuit of a test module (emergency stop, automatic discharge); design completed, build not carried out

PROJECTS

Precision LCR-meter (PCB v3)
MAR 2025 – JUN 2026

HELHA, Master's project, team of two

- Rethought the architecture of an auto-balanced bridge to target 0.5 %: ESP32 and external 12-bit AD7450 ADC, as the internal ADC lacked linearity
- Designed PCB v3 in KiCad: mixed-signal board (AD9850 DDS, INA128, PGA103, DAC8811) with a continuous ground plane and analog/digital separation
- Board assembled, analog chain validated on the oscilloscope up to the DAC (two routing defects worked around); firmware not carried out

Tracked Robot with 6-Axis Arm
FEB – JUN 2024

HELHA, Bachelor's project, team of three

- Designed the electronics: 3 KiCad PCBs (power, control, telemetry) with H-bridges, 12 V to 5 V DC/DC, battery cut-off and protections
- Defined the architecture (Raspberry Pi, ATmega, ESP32-CAM, SPI/I^2C/UART) and the web control; €260 (budget €300)

Game on Intel Cyclone V FPGA (VHDL)
OCT 2025

HELHA, Master's project, team of three

- Designed a complete game in VHDL: state machine, debouncing, collision detection
- Verified 8 modules with test benches: 201 logic elements, +13.6 ns setup slack at 50 MHz

CONTACT

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EDUCATION

Master's, Industrial Engineering (Electronics)
HELHA, Mons
2024 – 2026 · graduated

Bachelor's, Industrial Engineering
HELHA, Mons
2020 – 2024

SKILLS

EMBEDDED
C/C++ · FreeRTOS · STM32 · ESP32 · embedded Linux · UART · SPI · I^2C · CAN · BLE

HARDWARE
KiCad · Altium · AutoCAD · LTspice · EMI/EMC · IPC

FPGA
VHDL · Quartus Prime · Cyclone V

UAV
ArduPilot · MAVLink · LiDAR · ToF · EKF · DFT

TOOLS
Python · Bash · VBA · Git · MATLAB · oscilloscope

LANGUAGES

French	native
English	C1
Dutch	A1
Mandarin	A1

AVAILABILITY

Immediate
Driving licence B

SCAN Portfolio